



demand as the industry transitions from combustion engines. However, an oversupply of more common logic semiconductors and utility chips may occur with the influx of new factories. Additionally, political factors are causing market fragmentation, with governments imposing restrictions and tariffs on chips from rival countries, affecting supply chains and creating both opportunities and inefficiencies.

SG: Many fab projects, including those in Ohio and Arizona, are delayed. What are the reasons, and how might this affect investments?

CM: There are two primary reasons for the delays. First, the consumer electronics market, including PCs and smartphones, has been slow for the past year and a half, leading businesses to postpone wafer fab projects. Second, there is a shortage of skilled workers needed for new fab construction and operation, particularly in the US, where the number of new fabs has increased significantly. This shortage of specialised workers and the need for additional training are causing further delays.

SG: Regarding China, with numerous US sanctions on advanced semiconductor manufacturing technology and equipment, can China develop indigenous capabilities to produce sub-3-nanometre chips by 2030?

CM: Catching up in the chip industry is challenging due to its rapid advancement. TSMC currently leads in advanced fab manufacturing, while ASML is ahead in advanced lithography. By 2030, China might develop technologies similar to EUV for producing 3nm and 2nm chips, but their effectiveness in terms of yield and quality remains uncertain. Meanwhile, TSMC and ASML are likely to advance further, making

it increasingly difficult for China to catch up.

SG: Can China develop 2nm chips using their own equipment and technology by 2030?

CM: It is plausible that China could develop their own 2nm technology and manufacturing equipment by 2030. SMIC has been approximately five years behind TSMC. If TSMC starts producing 2nm chips in 2025, SMIC might reach 2nm by 2030, provided they maintain their current pace. However, developing new tools and processes will be crucial, and other regions may advance further technologically by then.

SG: If China were to develop a competitive mobile app processor and decided not to use, for instance, Qualcomm chips in their smartphones, what would be the implications?

CM: If China were to design, manufacture, and scale up production of smartphone processors with cutting-edge capabilities, they would need to avoid outdated technology. Currently, it is unlikely that China has the capacity to meet such demands with their existing technology.

SG: What would the impact be if this scenario occurred?

CM: Such a scenario would be significant. Chipmakers who primarily sell to Chinese brands would be particularly concerned. As a result, there is a shift towards moving production outside of China. For example, Apple plans to assemble a quarter of its smartphones in India starting next year. Non-Chinese chip design firms are also interested in expanding their presence outside of China. India, with its large smartphone market, offers opportunities for American, Taiwanese, and Japanese firms to collaborate with

Indian companies, thus reducing the market power of Chinese firms.

SG: Considering that approximately 76% of Indian smartphones are Chinese brands, what impact would it have on US semiconductor companies if Indian manufacturers chose MediaTek processors over Qualcomm due to trade tensions?

CM: This is a significant concern. US semiconductor companies like Qualcomm and Broadcom are mindful of these risks. They are working diligently to maintain strong relationships with Chinese firms and the government while also seeking to diversify their supply chains away from China to mitigate potential risks.

SG: Considering the current geopolitical situation, could India function as a strategic advantage for US semiconductor companies? For instance, if a company like Qualcomm collaborates with Indian firms to create an Indian brand of mobile phones using their processors, potentially capturing 10%-20% of the market share from Chinese brands, what is the feasibility of such a collaboration?

CM: This scenario is increasingly feasible due to growing geopolitical pressures and investments in the electronics ecosystem. US companies have a strong incentive to support the development of alternative products and brands in India. There is no reason an Indian-based smartphone company could not produce a world-class product, sourcing components from the US, Taiwan, Japan, and India, and compete effectively against Chinese brands. This opportunity extends beyond smartphones to other products where Chinese brands are also vulnerable, making it plausible to build competitive Indian products with the assistance of US semiconductor companies.